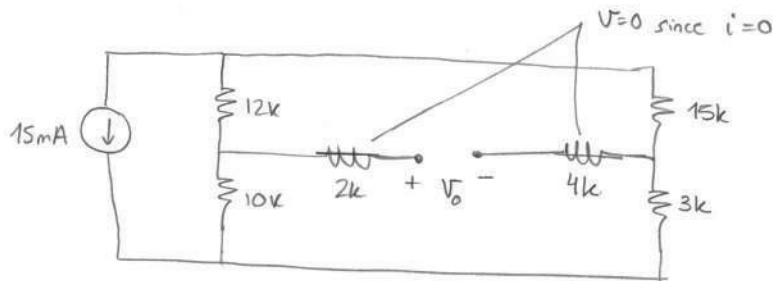
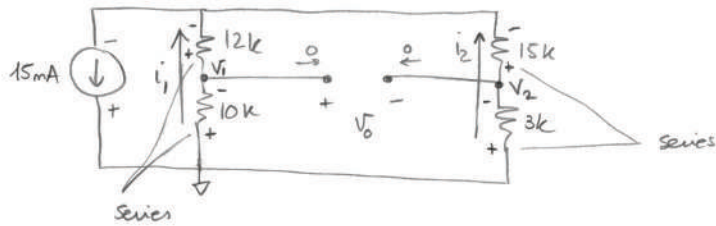


QUESTION 12



i)



$$i_1 = 15\text{mA} \frac{22\text{k} \parallel 18\text{k}}{22\text{k}} = 15\text{mA} \frac{9.9\text{k}}{22\text{k}} = 6.75\text{mA}$$

↑
current divider

$$i_2 = i - i_1 = 15\text{mA} - 6.75\text{mA} = 8.25\text{mA}$$

↑
KCL

$$V_1 + V_o - V_2 = 0 \Rightarrow V_o = V_2 - V_1, \text{ where } V_1 = V_{10k} = 10\text{k} \times i_1 = 67.5\text{V}$$

$$V_2 = V_{3k} = 3\text{k} \times i_2 = 24.75\text{V}$$

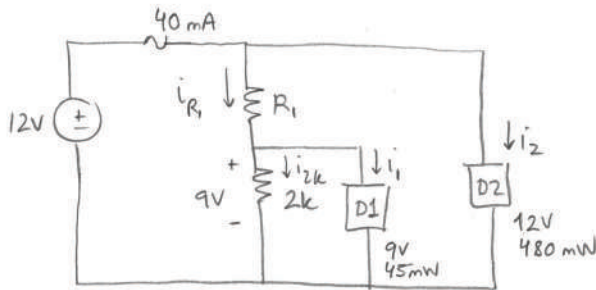
$$V_o = V_2 - V_1 = 24.75 - 67.5 = -42.75\text{V}$$

ii)

$$P_{15\text{mA}} = -V_{15\text{mA}} \times 15\text{mA} = -2.2275\text{W}$$

$$V_{15\text{mA}} = (12\text{k} + 10\text{k}) \times i_1 = 22\text{k} \times 6.75\text{mA} = 148.5\text{V}$$

• QUESTION 1b



i)

Calculate current in devices using details provided.

$$P = V \times i \Rightarrow i = \frac{P}{V} \Rightarrow i_1 = \frac{45 \text{ mW}}{9 \text{ V}} = 5 \text{ mA}$$

$$i_2 = \frac{480 \text{ mW}}{12 \text{ V}} = 40 \text{ mA}$$

Since D1 is parallel to 2k resistor

$$i_{2k} = \frac{9 \text{ V}}{2 \text{ k}} = 4.5 \text{ mA}$$

$$i_{\text{fuse}} = i_{R_1} + i_2 = 49.5 \text{ mA}$$

The fuse will burn out

Using KCL: $i_{R_1} = i_{2k} + i_1 = 4.5 \text{ mA} + 5 \text{ mA} = 9.5 \text{ mA}$

Also, using KVL: $V_{R_1} = 12 \text{ V} - 9 \text{ V} = 3 \text{ V} \Rightarrow \boxed{R_1 = \frac{V_{R_1}}{i_{R_1}} = \frac{3 \text{ V}}{9.5 \text{ mA}} = 315.79 \Omega}$

ii)

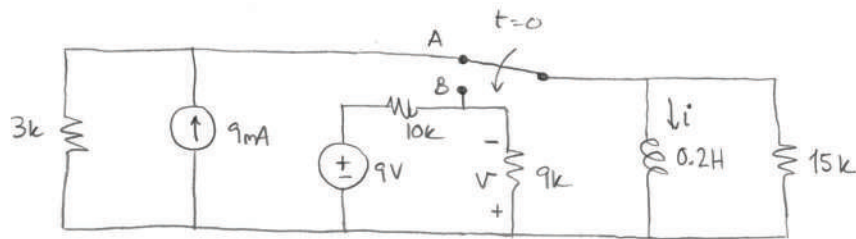
The fuse does not change current in middle branch and R_1 remains the same, so

$$\underset{\text{KCL}}{i_{\text{fuse}}} = i_{R_1} + i_2, \text{ where } i_{R_1} = 9.5 \text{ mA}$$

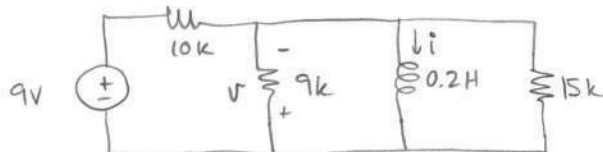
$$i_2 = i_{\text{fuse}} - i_{R_1} = 40 \text{ mA} - 9.5 \text{ mA} = 30.5 \text{ mA}$$

$$\boxed{P_2 = V_2 \times i_2 = 12 \text{ V} \times 30.5 \text{ mA} = 0.366 \text{ W}}$$

• QUESTION 22

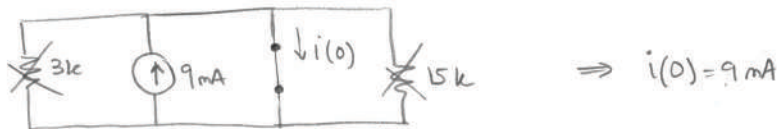


i) For $t > 0$, switch is in position B



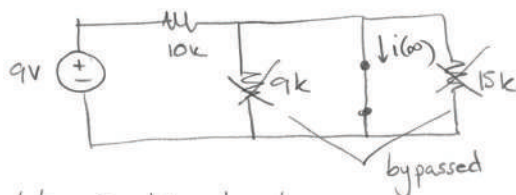
Step response: $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$, where $\tau = L/R$

* Calculate $i(0)$ for $t \rightarrow 0$ → switch in position A
L is short circuit (steady state)



$$\Rightarrow i(0) = 9 \text{ mA}$$

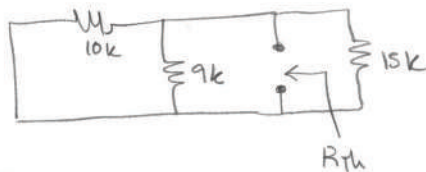
* Calculate $i(\infty)$ for $t \rightarrow \infty$ → switch in position B
L is short circuit (steady state)



$$i(\infty) = \frac{9V}{10k} = 0.9 \text{ mA}$$

* Calculate $\tau = L/R_{Th}$ for $t > 0$

• Calculate R_{Th} :
(method 1)



$$R_{Th} = 10k \parallel 9k \parallel 15k = 3.6 \text{ k}\Omega$$

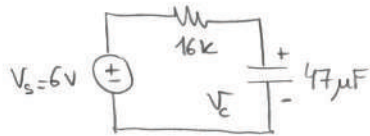
$$\tau = \frac{L}{R_{Th}} = \frac{0.2H}{3.6 \text{ k}\Omega} = 0.0555 \text{ ms} = 55.5 \mu\text{s}$$

* Substitute: $i(t) = 0.9 + [9 - 0.9]e^{-t/55.5 \times 10^{-6}} \text{ mA} = 0.9 + 8.1e^{-18000t} \text{ mA}$

ii) $V(t) = -V_L(t)$ where $V_L = L \frac{di}{dt} = 0.2 \frac{d}{dt} (0.9 + 8.1e^{-18000t}) \text{ m} =$
 $= 0.2 \times 8.1 \times (-18000) e^{-18000t} \text{ m} = -29.16 e^{-18000t} \text{ V}$
 $V(t) = 29.16 e^{-18000t} \text{ V}$

• QUESTION 2b

At $t > 0$ controller is open circuit and capacitor discharged (i.e., $V_c(0) = 0V$)

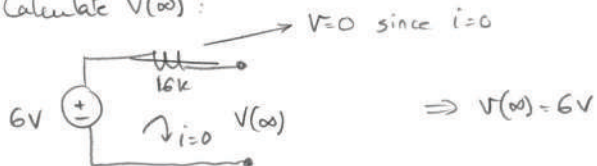


Step response:

$$V_c(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}, \text{ where } \tau = RC$$

* $V_c(0) = 0$

* Calculate $V(\infty)$:



$$\Rightarrow V(\infty) = 6V$$

* Calculate τ :

$$\tau = RC = 16k \times 47\mu F = 0.752s$$

* Substitute:

$$V_c(t) = 6 + [0 - 6]e^{-t/0.752} = 6(1 - e^{-1.33t})V$$

Make $V_c = 0.8V_s = 0.8 \times 6 = 4.8V$

$$6(1 - e^{-1.33t}) = 4.8V$$

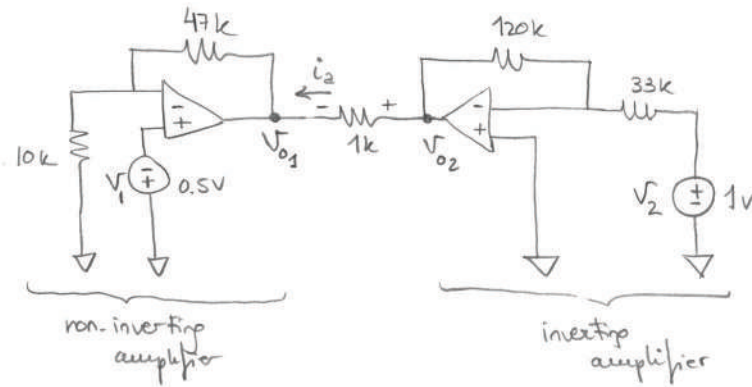
$$-e^{-1.33t} = 4.8/6 - 1$$

$$+e^{-1.33t} = +0.2 \Rightarrow \ln(e^{-1.33t}) = \ln(0.2)$$

$$-1.33t = -1.61$$

$$\boxed{T = \frac{+1.61}{+1.33} = 1.21s}$$

QUESTION 32



i)

$$i_2 = \frac{V_{o2} - V_{o1}}{1k}, \text{ where}$$

$$V_{o1} = \left(1 + \frac{47k}{10k}\right) (-0.5V) = 5.7 (-0.5) = -2.85V$$

↑
non-inverting

$$V_{o2} = -\frac{120k}{33k} \times 1V = -3.64V$$

↑
inverting

$$i_2 = \frac{V_{o2} - V_{o1}}{1k} = \frac{-3.64 - (-2.85)}{1k} = -0.79 \text{ mA}$$

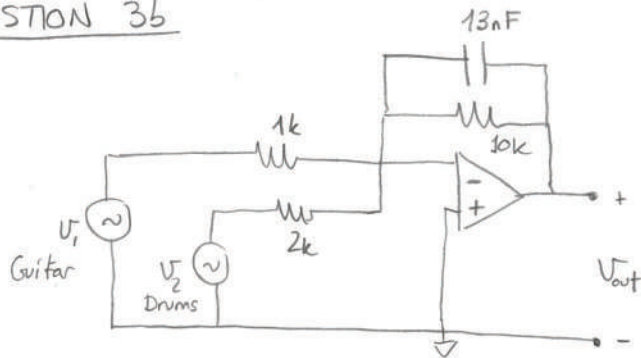
ii)

$$i_2 = 0 \text{ if } V_{o1} = V_{o2}$$

$$V_{o2} = -\frac{120k}{33k} V_2 = -3.64 V_2 = -2.85V \Rightarrow \boxed{V_2 = \frac{+2.85}{+3.64} = 0.783V}$$

↑
inverting

• QUESTION 3b

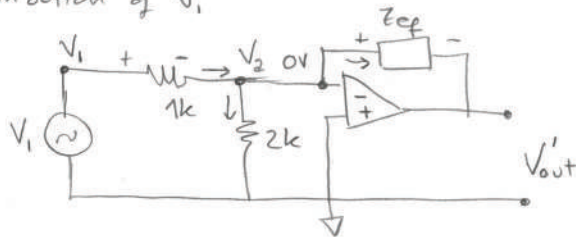


$$V_1 = 0.24 \cos(400\pi t) \text{ V}$$

$$V_2 = 0.21 \cos(120\pi t) \text{ V}$$

i) We need to use superposition since we have different frequencies

* Contribution of V_1



* Convert to phasor domain

$$V_1(t) = 0.24 \cos(400\pi t) \text{ V}$$

$$\Rightarrow V_1 = 0.24 \angle 0^\circ \text{ V}$$

$$Z_c = -j \frac{1}{\omega c} = -j \frac{1}{400\pi \times 13 \times 10^{-9}} = -j 61213.44 \, \Omega$$

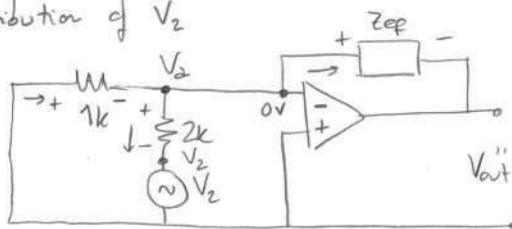
$$\text{KCL @ } V_2: \frac{V_1 - V_2}{1k} - \frac{V_2 - 0}{2k} - \frac{V_2 - V'_{out}}{Z_{eq}} = 0$$

$$V_2 = 0 \text{ V (negative feedback)} \Rightarrow \frac{V_1}{1k} + \frac{V'_{out}}{Z_{eq}} = 0 \Rightarrow V'_{out} = -\frac{Z_{eq}}{1k} V_1$$

$$\text{where } Z_{eq} = 10k \parallel (-j 61213.44) = \frac{10000 \times (-j 61213.44)}{10000 - j 61213.44} = \frac{61213.44 \times 10^4 \angle -90^\circ}{62024.88 \angle -80.72^\circ} = 9869.18 \angle -9.28^\circ$$

$$V'_{out} = -\frac{Z_{eq}}{1k} V_1 = \frac{9869.18 \angle 180 - 9.28^\circ \times 0.24 \angle 0^\circ}{1000} = 2.369 \angle 170.72^\circ \text{ V}$$

* Contribution of V_2



* Convert to phasor domain

$$V_2(t) = 0.21 \cos(120\pi t) \text{ V} \Rightarrow V_2 = 0.21 \angle 0^\circ$$

$$Z_c = -j \frac{1}{\omega c} = -j \frac{1}{120\pi \times 13 \times 10^{-9}} = -j 204044.8 \, \Omega$$

$$\text{KCL @ } V_2: \frac{0 - V_2}{2k} - \frac{V_2 - 0}{10k} - \frac{V_2 - V''_{out}}{Z_{eq}} = 0$$

$$V_2 = 0 \Rightarrow \frac{V_2}{2000} + \frac{V''_{out}}{Z_{eq}} = 0 \Rightarrow V''_{out} = -\frac{Z_{eq}}{2k} V_2$$

$$\text{where } Z_{eq} = 10k \parallel (-j 204044.8) = \frac{204044.8 \times 10^4 \angle -90^\circ}{10000 - j 204044.8} = \frac{204044.8 \times 10^4 \angle -90^\circ}{204289.7 \angle -87.19^\circ} = 0.9999 \times 10^4 \angle -2.81^\circ \, \Omega$$

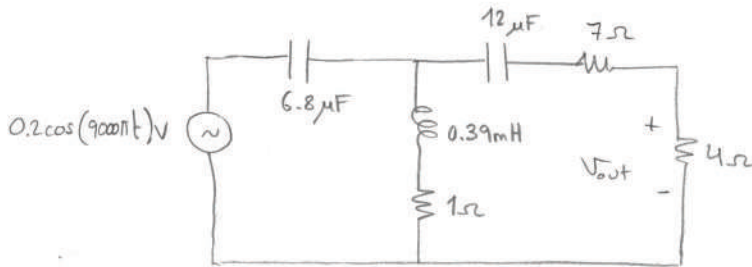
$$V''_{out} = \frac{-9999 \angle -2.81^\circ}{2000} \times 0.21 \angle 0^\circ = 1.05 \angle 177.19^\circ \text{ V}$$

* Add in time domain $V_{out} = 2.369 \cos(400\pi t + 170.72^\circ) + 1.05 \cos(120\pi t + 177.19^\circ) \text{ V}$

QUESTION 4

Alternative

- source transformation
+ current divider



i)

First, convert to phasor domain:

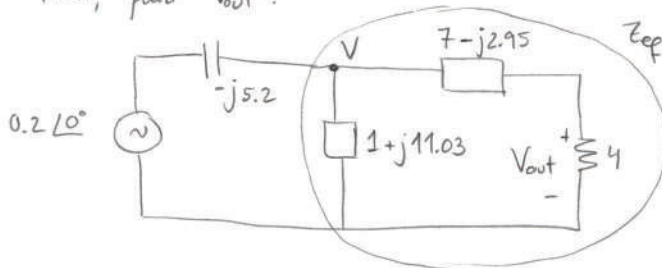
$$V_{in} = 0.2 \cos(9000\pi t) \text{ V} \Rightarrow V_{in} = 0.2 \angle 0^\circ$$

$$Z_{C1} = Z_{6.8\mu F} = -j \frac{1}{\omega C_1} = -j \frac{1}{9000\pi \times 6.8 \times 10^{-6}} = -j 5.2 \Omega$$

$$Z_{C2} = Z_{12\mu F} = -j \frac{1}{\omega C_2} = -j \frac{1}{9000\pi \times 12 \times 10^{-6}} = -j 2.95 \Omega$$

$$Z_L = j\omega L = j 9000\pi \times 0.39 \times 10^{-3} = j 11.03 \Omega$$

Then, find V_{out} :



$$Z_{eff} = [4 + 7 - j 2.95] \parallel (1 + j 11.03) = (11 - j 2.95) \parallel (1 + j 11.03) = \frac{(11 - j 2.95)(1 + j 11.03)}{11 - j 2.95 + 1 + j 11.03} = \frac{11.39 \angle -15.01^\circ \times 11.08 \angle 84.82^\circ}{12 + j 8.08} = \frac{126.2 \angle 69.81^\circ}{14.47 \angle 33.95^\circ} = 8.72 \angle 35.86^\circ \Omega = 7.07 + j 5.11 \Omega$$

$$V = 0.2 \angle 0^\circ \text{ V} \quad \text{Voltage divider} \quad \frac{Z_{eff}}{-j 5.2 + Z_{eff}} = \frac{0.2 \angle 0^\circ \times 8.72 \angle 35.86^\circ}{-j 5.2 + 7.07 + j 5.11} = \frac{1.744 \angle 35.86^\circ}{7.07 - j 0.09} = \frac{1.744 \angle 35.86^\circ}{7.07 \angle -0.73^\circ} = 0.247 \angle 36.59^\circ \text{ V}$$

$$V_{out} = V \quad \text{Voltage divider} \quad \frac{4}{4 + 7 - j 2.95} = \frac{0.247 \angle 36.59^\circ \times 4}{11 - j 2.95} = \frac{0.988 \angle 36.59^\circ}{11.39 \angle -15.01^\circ} = 0.0867 \angle 51.6^\circ \text{ V}$$

Finally, convert back to time domain:

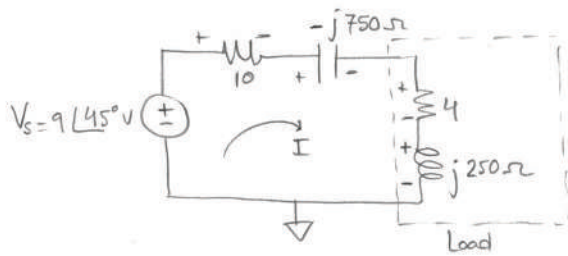
$$|V_{out}(t) = 0.0867 \cos(9000\pi t + 51.6^\circ) \text{ V}|$$

ii)

$$P_{4\Omega} = \frac{1}{2} \frac{|V_{out}|^2}{4} = \frac{1}{2} \frac{0.0867^2}{4} = 0.94 \text{ mW}$$

$V_{4\Omega} = V_{out}$

• QUESTION 52



$$P_{\text{speaker}} = P_{4\Omega} + P_{j250}$$

0 (average power in inductor is 0W)

Using KVL:

$$-9\angle45^\circ + 10I + (-j750)I + 4I + j250I = 0$$

$$I = \frac{9\angle45^\circ}{10 - j750 + 4 + j250} = \frac{9\angle45^\circ}{14 - j500} = \frac{9\angle45^\circ}{500.2\angle-88.4^\circ} = 0.018\angle133.4^\circ \text{ A}$$

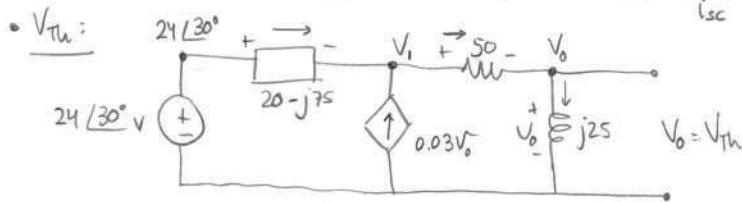
$$P_{\text{speaker}} = P_{4\Omega} = \boxed{\frac{1}{2} * 4 * |I|^2 = 2 * |0.018|^2 = 648\mu\text{W}}$$

• QUESTION 5b



i) MAPT: $R_{load} = |Z_{Th}|$

★ Calculate Z_{Th} (method 2): $Z_{Th} = \frac{V_{Th}}{I_{sc}}$ → also needed to calculate P_{max}



Another option:
2 source transf.
+ voltage divider

KCL@ V_1 : $\frac{24\angle 30^\circ - V_1}{20 - j75} + 0.03V_o - \frac{V_1 - V_o}{50} = 0$ ①

KCL@ V_o : $\frac{V_1 - V_o}{50} = \frac{V_o}{j25} \Rightarrow 0.02V_1 - 0.02V_o = -j0.04V_o \Rightarrow 0.02V_1 - (0.02 - j0.04)V_o = 0$ ②

Solve for V_o :

$$\frac{24\angle 30^\circ}{77.62\angle -75.07^\circ} - \left(\frac{1}{77.62\angle -75.07^\circ} + \frac{1}{50} \right) V_1 + (0.03 + 0.02)V_o = 0$$

$$0.31\angle 105.07^\circ - (0.0129\angle 75.07^\circ + 0.02)V_1 + 0.05V_o = 0$$

$$(0.023 + j0.012)V_1 - 0.05V_o = 0.31\angle 105.07^\circ$$

$$\left. \begin{aligned} (0.023 + j0.012)V_1 - 0.05V_o &= 0.31\angle 105.07^\circ \quad (\times (-0.02)) \\ 0.02V_1 - (0.02 - j0.04)V_o &= 0 \quad (\times (0.023 + j0.012)) \end{aligned} \right\}$$

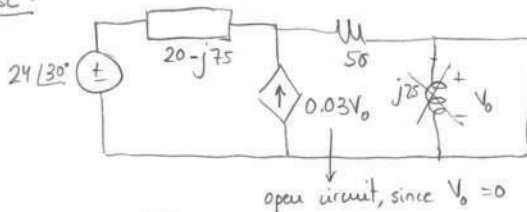
$$-0.02(0.023 + j0.012)V_1 + 0.001V_o = 0.0062\angle 105.07^\circ - 180^\circ$$

$$0.02(0.023 + j0.012)V_1 - (0.02 - j0.04)(0.023 + j0.012)V_o = 0$$

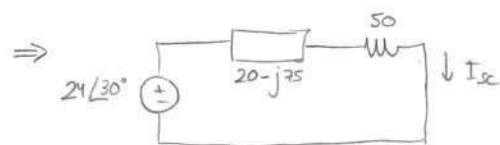
$$(0.001 - 0.00046 - j0.00024 + j0.00092 - 0.00048)V_o = 0.0062\angle -74.93^\circ$$

$$V_o = \frac{0.0062\angle -74.93^\circ}{0.00006 + j0.00068} = \frac{0.0062\angle -74.93^\circ}{0.000683\angle 84.96^\circ} = 9.077\angle -159.89^\circ V = V_{Th}$$

• I_{sc} :

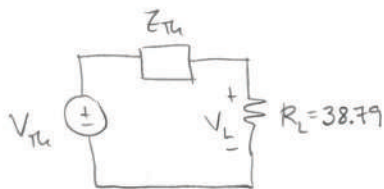


$$I_{sc} = \frac{24\angle 30^\circ}{70 - j75} = \frac{24\angle 30^\circ}{102.59\angle -46.97^\circ} = 0.234\angle 76.97^\circ$$



$$\Rightarrow Z_{Th} = \frac{9.077\angle -159.89^\circ}{0.234\angle 76.97^\circ} = 38.79\angle 123.14^\circ \Omega$$

$$R_{load} = |Z_{Th}| = 38.79 \Omega$$



$$P_{max} = \frac{1}{2} \frac{|V_L|^2}{R_L}$$

where $V_{th} = 9.077 \angle -159.89^\circ \text{ V}$

$$Z_{th} = 38.79 \angle 123.14^\circ \Omega = -21.21 + j32.48$$

$$R_L = 38.79 \Omega$$

Calculate V_L :

$$V_L = V_{th} \frac{R_L}{Z_{th} + R_L} = 9.077 \angle -159.89^\circ \frac{38.79}{38.79 - 21.21 + j32.48} =$$

$$= \frac{352.097 \angle -159.89^\circ}{17.58 + j32.48} = \frac{352.097 \angle -159.89^\circ}{36.93 \angle 61.58^\circ} = 9.534 \angle 138.53^\circ \text{ V}$$

$$P_{max} = \frac{1}{2} \frac{9.531^2}{38.79} = 1.172 \text{ W}$$

ii) $P_{max} = \frac{|V_{th}|^2}{8R_{th}} = \frac{9.077^2}{8 \times (-21.21)} = -0.486 \text{ W}$

Note: Since we have a negative resistance, we would end up with a negative power, which does not make sense for a load (we would like students to mention this, but we are not penalising them if they do not mention it)